

HIGHER TIER ONLY QUESTIONS

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
3	(a)			award (1) each for up to two of following • speeds up a chemical reaction • lowers activation energy • not used up during the reaction doesn't take part in the reaction - neutral answer	2			2		
	(b)	(i)		award (1) for any <u>comparison</u> of active ranges • A works in pH range of 0.5-4.5 <u>and</u> B works in pH range 3-8 • A works at a lower pH range / B works at a higher pH range • A works over a narrower pH range / B works over a wider pH range award (1) for <u>comparison</u> of optimum pH e.g. • A works best at pH 2 <u>and</u> B works best at pH 5.5 • A works best at a lower pH / B works best at a higher pH award (1) for <u>comparison</u> of activity at given points • both have the same activity at their optimum pH • both have the same activity at pH 3.75 up to maximum (2)			2	2		
		(ii)		curve drawn rising from pH 5 then falling to pH 9 (1) peak at pH 7 (1)			2	2		

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(c)			activity increases up to optimum temperature (1) decreases after optimum temperature (1) rate of decrease is more rapid than rate of increase (1) reference to denaturing / lock and key - neutral answers			3	3		
				Question 3 total	2	0	7	9	0	0